VikasSetu — Aamchi Shala

Branch Changes & Implementation Summary: Member 2

Branch Name:	`sarvesh-observation-management`	**Date / Time:**	18 October 2026, 15:35 IST
Author:	SarveshVeshi (sarveshveshi@gmail.com)	**Module:**	Member 2: Observation Management
Base Branch:	`main (Zero modifications)`	**Commit Hash:**	`6447bf4`

1. High-Level Summary

This branch implements the complete **Member 2: Observation Management (Form & Data Capture)** module for the VikasSetu preschool support platform. The system connects the frontend React 19 application directly to Member 1's Python FastAPI backend and SQLite database (backend/vikassetu.db), enabling teachers to record, validate, edit, delete, and persist classroom observations for children aged 3–4 across five learning domains. All data persists permanently across page refreshes and automatically drives derived child scores, strengths, and needs.

- **Dynamic Child Population:** Real children profiles are fetched dynamically from the FastAPI backend (GET /api/children).
- **Full Form Capture:** Captures Child ID, Observation Date (with date picker), Learning Area/Domain, Activity, Rating/Score (1–3), and Notes.
- **Input Validation:** Client-side validation enforces non-empty fields, valid date formatting, category constraints, and non-empty notes (with a 500-character limit counter) with inline error alerts.
- **Backend Error Resilience:** Clear error alert banner surfaces HTTP 400/422/500 errors from FastAPI (such as activity-domain mismatch).
- **Duplicate Submission Guard:** Interactive loading spinners and disabled button states prevent duplicate submissions.
- **Saved Observations & In-Place Editing:** Real-time list of saved observations fetched from GET /api/observations with child/domain filters and baseline toggles, plus Edit (PUT) and Delete (DELETE) modals with automatic child score recalculation.
- **100% Verification:** 10/10 automated backend integration tests, 13/13 Vitest tests, and zero TypeScript / build errors.

2. File-by-File Breakdown

File Path	Status	Description of Changes & Rationale
`src/lib/api.ts`	**NEW FILE**	Created centralized API client connecting frontend to FastAPI backend. Provides typed methods for fetchChildren, fetchChild, fetchObservations, createObservation, updateObservation, deleteObservation, and fetchActivities. Implements robust error extraction parsing FastAPI 422 validation detail arrays and string messages.
`src/routes/progress.tsx`	**MODIFIED**	Replaced mock placeholder with full Member 2 Observation Management system. Implements dynamic child selector, observation date picker, domain selector, activity selector, 1–3 score selector, and required notes textarea with 500-char counter. Adds client-side validation with inline error messages, backend error banners, loading spinners, and a Saved Observations section with child/domain filters, baseline toggles, Edit modal (PUT), and Delete modal (DELETE). Re-fetches child profile on changes to keep scores in sync.

`src/lib/data.ts`	**MODIFIED**	Enhanced TypeScript interfaces without breaking existing contracts. Added isBaseline?: boolean to Observation and observedDomains?: DomainId[] to Child. Applied non-null assertion on domainOf helper to eliminate undefined indexed-access compiler errors.
`src/lib/store.tsx`	**MODIFIED**	Added non-breaking state synchronization helpers: syncChild, syncObservation, removeObservationFromStore, setChildren, and setObservations. Allows observation updates to instantly propagate to the shared global store so Member 3 & 4 pages stay updated.
`src/test/observation-management.test.tsx`	**NEW FILE**	Created comprehensive Vitest unit test suite validating API client operations: verifies GET children, POST create observation, PUT update observation, DELETE observation, FastAPI 422 unprocessable entity handling, and 404 not found handling.

Deleted Files: None (Zero deletions; existing team files preserved intact).

3. How Teammates Can Test & Verify

1. Pull and Switch to Branch:
git fetch origin
git checkout sarvesh-observation-management
git pull origin sarvesh-observation-management

2. Start Backend (Terminal 1):
cd backend
.\venv\Scripts\uvicorn.exe app.main:app --port 8000 --reload
Verify: Swagger docs live at http://localhost:8000/docs

3. Start Frontend (Terminal 2):
npm install --legacy-peer-deps
npm run dev -- --port 5173
Open: http://localhost:5173/progress in your browser

4. Interactive Verification Steps:
• **Validation Check:** Click 'Save Observation' without selecting fields to see inline validation errors.
• **Create Observation:** Select a child (e.g. Aarav), observation date, learning area, activity, score (1–3), and notes. Click 'Save Observation' to verify success and SQLite persistence.
• **Saved Observations & Filter:** Scroll to 'Saved Observations' below the form. Test filtering by child and learning area.
• **Edit & Delete:** Click the Pencil icon to edit an observation or Trash icon to delete it. Check that child scores automatically recalculate.
• **Persistence:** Refresh the browser (F5) to confirm all observations and updated scores persist from SQLite.

5. Run Test Suite:
npm run test (All 13 Vitest tests pass)
npx tsc --noEmit (Zero TypeScript compilation errors)
npm run build (Production bundle builds cleanly)